

# DH<sup>2</sup>A-KT: Diagnostic Heterogeneous Hypergraph and Agentic Knowledge Tracing with Leakage-Controlled Graph Construction

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## Abstract

Graph-augmented knowledge tracing (KT) often lacks leakage audits of its auxiliary graphs and rarely exposes frozen predictions to a gated explanation layer. We present DH<sup>2</sup>A-KT, a two-tier system that is tractable on a single consumer GPU. **Tier 1 (DH<sup>2</sup>-KT)** reuses the leakage-controlled, train-only concept-graph protocol of companion paper P0, groups audited prerequisite edges into *concept-prerequisite chain hyperedges* (the quantitative core evaluated here), and extends P0’s diagnostics with hyperedge-level leakage checks. Discussion-thread and teacher-intervention types remain data-blocked; session+Hint hyperedges and a limited observational hint ATE are demonstrated on FoundationalASSIST (not on XES3G5M). **Tier 2 (AgentDG-KT)** is a small-scale pilot: Diagnostician, Critic, and Tutor agents read Tier 1’s *frozen* outputs. On XES3G5M, DH<sup>2</sup>-KT v2 (chain

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hyperedges, graph-only interaction) achieves mean test AUC 0.752 versus reused GKT 0.834 under matched budget, while passing hypergraph manipulation checks on all three folds ( $\Delta\text{AUC} = 0.038\text{--}0.047$  at  $p=0.9$  node-drop); P0 GKT also passes destruction at the same anchor with a larger mean drop, so reliance is not unique to  $\text{DH}^2\text{-KT}$ . Junyi is used only for structural ground-truth overlap (not full model training). A 500-sample Tier 2 pilot with Qwen2.5-7B (Ollama) yields Critic flag rates of 12.4% under a strict prompt and 0% after probability-format calibration; grounded vs. independent-generation explanations give flag rates 0.026/0.768 and mean ID-anchored KC-Jaccard 0.152/0.142. A 40-sample dual-rater study rates faithfulness  $4.01 \pm 1.07$  and usefulness  $3.53 \pm 0.71$  (Likert 1–5). FoundationalASSIST supplies session+Hint hyperedges and a limited observational hint ATE (Section 7.5).

*Keywords:* Knowledge tracing, hypergraph neural networks, heterogeneous graphs, large language model agents, causal inference, data leakage, educational data mining

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## 1. Introduction

Knowledge tracing (KT) estimates a learner’s latent mastery of knowledge components (KCs) from interaction history, supporting adaptive practice, intelligent tutoring, and curriculum recommendation. The field has progressed from probabilistic and sequence models [1] to graph-augmented models that propagate information over auxiliary concept graphs [2, 3], and, most recently, toward hypergraph representations that capture higher-order relations among knowledge components [4, 5, 6] and dynamic, temporally-

evolving graph structure [7].

This paper builds on – rather than replacing – [8]’s leakage-control protocol. Empirically, Tier 1 evaluates *concept-prerequisite* chain hyperedges under a graph-only encoder and a manipulation check. Discussion-thread and teacher-intervention hyperedges remain data-blocked; only session+Hint hyperedges and a limited observational hint ATE are demonstrated on FoundationalASSIST (weak propensity overlap; Sections 4.1–4.4). Section 4.2 extends P0’s diagnostics to the hyperedge setting. Tier 2 (Section 5) is a deliberately-scoped pilot showing that an LLM-agent layer can read Tier 1’s frozen output without reopening the leakage question. On XES3G5M, graph-only interaction trades  $\approx 8.2$  AUC points versus GKT for a passing manipulation check; Tier 2 faithfulness at pilot scale is sensitive to Critic prompt design but reaches 0% flag rate after calibration (Section 6.3). Fig. 1 summarizes the two-tier pipeline.

### 1.1. Motivation

Three concrete gaps motivate this work:

- **Higher-order structure is under-audited in graph-KT.** Hypergraph-KT models [4, 5, 6] group multiple knowledge components into a single hyperedge, yet published systems rarely audit whether those hyperedges were constructed without leakage, and few expose an interface for entities beyond KCs (hints, teachers, forums, video) when such logs become available.
- **Graph provenance is rarely audited.** [8] shows that pooling a knowledge-component graph from the full interaction log before a learner-

based train/test split lets held-out information reach a graph-augmented KT model even when the sequence optimizer itself trains only on the training fold. [8]’s protocol audits this channel for *pairwise* prerequisite/similarity edges; no published audit protocol currently covers *hyperedges*.

- **Causal claims about pedagogical intervention are rare.** Most graph-KT models report predictive accuracy only, without attempting to separate a hint or teacher action’s causal effect on subsequent correctness from confounded correlation — motivating an interface for ATE estimation even when public logs do not yet support reporting numbers.

## 1.2. Research Questions

This work is motivated by a broader question: *how can a heterogeneous hypergraph – with student, exercise, concept, teacher, hint, forum, discussion, and video as node types (Section 3) – improve knowledge tracing beyond pairwise student–concept–exercise graphs?* We decompose this into five questions answered with evidence below, plus one this paper does not answer:

- **RQ1** (§6.3): Does a leakage-controlled concept-prerequisite hypergraph yield competitive KT accuracy under a graph-only encoder?
- **RQ2** (§7.2): Does the resulting model demonstrably rely on the constructed hypergraph, and is that reliance unique relative to reused graph-KT baselines?
- **RQ3** (§7.3): Does concept-prerequisite hyperedge construction preserve structural fidelity with an expert-curated prerequisite DAG?

- **RQ4** (§7.4): Can a Critic-gated LLM-agent layer read frozen Tier 1 predictions and produce faithful, useful explanations at pilot scale?
- **RQ5** (§7.5): Can session+Hint hyperedges and a hint ATE be instantiated on a hint-bearing corpus, and what do propensity diagnostics reveal about identification?

Interactions involving *teacher*, *forum/discussion*, and *video* nodes are defined in the schema (Section 3) but remain data-blocked on every corpus we use (Section 8.4): this paper establishes an interface for those entities, not empirical evidence of how they improve KT.

### 1.3. Contributions

- **C1.** A hypergraph construction procedure (Section 4.1) that groups [8]’s audited, acyclic prerequisite edges into concept-prerequisite chain hyperedges (*evaluated quantitatively* on XES3G5M/Junyi), plus session hyperedges with Hint members on FoundationalASSIST [9] (Section 7.5); discussion-thread and teacher-intervention types remain data-blocked (Section 8.4).
- **C2.** A hyperedge-level extension of [8]’s four-diagnostic leakage taxonomy, adding three classes – group-membership, cross-modal, and intervention-timing leakage – that a pairwise audit structurally cannot detect (Section 4.2).
- **C3.** A causal-effect layer (Section 4.4) implementing inverse-propensity ATE estimation; on FoundationalASSIST we report a *limited* observational hint ATE (next-step correctness) with propensity diagnostics

(Section 7.5), while XES3G5M still has no synchronized hint treatment for ATE.

- **C4.** Direct reuse of [8]’s published, leakage-controlled baseline results (BKT, DKT, AKT, simpleKT, GKT, GIKT, SKT, DyGKT, DGEKT) as the Tier 1 comparison column under matched protocol and split seeds, avoiding redundant baseline retraining.
- **C5.** A Tier 2 pilot (AgentDG-KT, Section 5) showing that a Critic-gated LLM interpretability loop over frozen Tier 1 outputs is tractable at  $n=500$  on a consumer GPU, plus a dual-rater Likert study ( $n=40$ ) of faithfulness/usefulness (Table 9) — scoped as pilot evidence, not a full classroom RCT.
- **C6.** Empirical evidence on XES3G5M that (i) a graph-only  $DH^2$ -KT backbone passes hypergraph manipulation checks on all three folds ( $\Delta AUC= 0.038\text{--}0.047$ ) while remaining below reused GKT AUC (mean  $0.752\pm 0.002$  vs.  $0.834\pm 0.001$ ), noting that P0 GKT *also* passes destruction at  $p=0.9$  with a larger mean drop, and (ii) the local 7B Tier 2 pilot reaches Critic flag rates of 0.026 (grounded) vs. 0.768 (independent generation) with mean ID-anchored KC-Jaccard 0.152/0.142 (Section 6.3).

## 2. Related Work

### 2.1. Graph- and hypergraph-based knowledge tracing

Early graph-augmented KT models propagate information over a single, pairwise concept graph: [2] layer a graph neural network over a knowledge-component graph on top of a sequence model, and [3] let exercises and con-

cepts recursively re-rank one another through a bipartite exercise-concept graph. [7] extends this line to a temporally-evolving graph that updates edges as new interactions arrive rather than freezing the graph after construction. A second line moves from pairwise edges to hyperedges spanning more than two knowledge components at once: [4] proposes a temporal hypergraph memory network, [5] adds a dual-gated mechanism controlling how much a hyperedge’s aggregated signal influences a given prediction, and [6] replaces graph convolution with a hypergraph transformer. A third line adds heterogeneity in what a node *represents* rather than how many nodes an edge spans: [10] augments the concept graph with psychological-factor nodes, and [11] builds a heterogeneous graph directly over learner-question interactions. Across all three lines, every published hyperedge or heterogeneous node we are aware of is still drawn from a closed set of KC-, exercise-, or learner-adjacent entities; none represent a Teacher intervention, Hint content, Forum discussion, or instructional Video as a first-class member inside the same hyperedge as the knowledge components it plausibly affects – the interface gap C1 (Section 4.1) targets, while this paper’s quantitative core evaluates concept-prerequisite chains.

## 2.2. LLM agents in education and graph-augmented LLM agents

[12] surveys the growing use of LLM agents across tutoring, feedback generation, and curriculum planning in education, and [13] instantiates a multi-agent LLM pipeline that simulates computerized adaptive testing end to end, with agents standing in for both the test administrator and the examinee. Outside education, [14] surveys how LLM agents can read from and write to an external graph as working memory, and [15] applies a graph-

retrieval-augmented-generation pattern to recommend personalized learning paths over a dual knowledge-structure graph. Closest in scope to our Tier 2 pilot is [16], which aligns LLM-generated mastery reports with implicit DLKT states (IKT-EIA) and evaluates explanation faithfulness with a corpus-wide KC-Jaccard metric, an independent-generation negative control, and paired predictive / faithfulness ablations on three large KT benchmarks. AgentDG-KT (Section 5) complements that line rather than competing head-to-head on the same datasets: we audit the auxiliary graph for train/test leakage before explanation, gate rationales with a mandatory Critic over a *frozen* Tier 1  $P(\text{correct})$ , and measure ID-anchored KC-Jaccard plus an independent-generation control adapted to our numeric-ID schema (XES3G5M lacks a synonym KC vocabulary in-repo). We adopt the frozen-core-plus-pilot-agent split specifically because a full multi-agent LLM pipeline that re-infers knowledge state end to end over millions of interactions is not tractable on a single consumer GPU (Section 5), not because we believe agentic re-inference is undesirable in principle.

### 2.3. Leakage-controlled evaluation in knowledge tracing

[8] is the companion paper this work builds on and reuses directly (Section 6.1); we summarize its protocol here so the rest of this paper is self-contained. [8] observes that a knowledge-component graph is often constructed by pooling co-occurrence or similarity statistics over the *entire* interaction log before a learner-based train/test split is applied, which lets held-out information reach a graph-augmented KT model through the graph channel even when the sequence model itself trains only on the training fold. Its audit protocol reports four scalar diagnostics computed on a train-



only-constructed graph – an edge-composition-ratio flag ( $\text{ECR}^{\text{flag}}$ ), an edge-composition overlap ratio ( $\text{ECR}^{\text{overlap}}$ ), an edge-outcome correlation ( $|\rho|$ ), and a train/test boundary-mismatch rate (TBMR) – alongside a DAG audit with cycle pruning, a cold-start stratification of knowledge components into four frequency strata, a DAG Disruption Rate (DDR) manipulation check under five structured augmentation operators, and a ground-truth cross-validation against an expert-curated prerequisite DAG on Junyi Academy. Across XES3G5M, ASSISTments 2012, and Junyi Academy, [8] reports that train-only graph construction shifts baseline AUC by at most 0.003 relative to full-log construction – evidence that leakage control, in their setting, does not by itself manufacture the reported gains, and a result we rely on directly when interpreting any Tier 1 gain in this paper (Section 6.1). Section 4.2 extends this four-diagnostic protocol from pairwise edges to heterogeneous hyperedges.

#### *2.4. Causal inference in educational data mining*

Causal analysis in education has mostly targeted two separate questions: estimating the effect of a known treatment, and discovering an unknown causal structure among skills. [17] addresses the first: given historical course-enrollment and grade data, they estimate the average treatment effect of taking one course before another using matching and regression – directly analogous in spirit to the propensity-score estimator we use for hint/teacher-intervention effects (Section 4.4), though on a coarser course-level treatment rather than a step-level pedagogical action. [18] addresses the second: they learn a latent causal ordering and dependency structure among knowledge-component skills end to end inside a modified DKT model, evaluated on the

NeurIPS 2022 Challenge on Causal Insights for Learning Paths in Education. [19] sits between the two, using a causal-relations-constrained student model to make KT predictions more interpretable without a separate treatment-effect estimation step. Our causal-effect layer (Section 4.4) is closest in spirit to [17] – a treatment-effect estimator rather than a causal-discovery method – but applies it at the level of individual hint/teacher-intervention hyperedges rather than curriculum-level course sequencing, and explicitly reports the propensity-overlap diagnostic that [17] does not discuss, following the identification-assumption caveat in Section 8.4.

### 3. Preliminaries

Let  $\mathcal{D} = \{(u, t, q, c, y)\}$  be a KT interaction log with learner  $u$ , timestamp  $t$ , item  $q$ , knowledge component  $c$ , and correctness label  $y \in \{0, 1\}$ , following [8]’s notation. A learner-based temporal split  $\mathcal{F} = (\mathcal{F}_{\text{train}}, \mathcal{F}_{\text{val}}, \mathcal{F}_{\text{test}})$  assigns each learner to exactly one fold (ratios 0.7/0.1/0.2, three folds with split seeds 42/43/44, matching [8]’s protocol so results remain directly comparable, see Section 6.1).

#### 3.1. Self-contained summary of the reused P0 protocol

Because [8] is under review at the time of writing, we summarize the reused pieces so this manuscript can be checked without reading the companion paper in full. **Construction.** For each learner-based fold, prerequisite edges  $E_{\text{pre}}^f$  and similarity edges  $E_{\text{sim}}^f$  are built from *train-fold interactions only*, then audited for cycles (DAG pruning). **Four pairwise diagnostics** (reported on the train-only graph against held-out learners): edge-composition-ratio flag  $\text{ECR}^{\text{flag}}$ , overlap  $\text{ECR}^{\text{overlap}}$ , edge–outcome correlation

$|\rho|$ , and train/test boundary-mismatch rate TBMR. Across XES3G5M, ASSISTments 2012, and Junyi, [8] finds that switching from full-log to train-only graph construction changes baseline AUC by at most 0.003 — evidence that leakage control alone does not manufacture large gains. **Reuse artefacts.**

We pin the companion repository to commit `5e3d6a01c32a649f1655ac8712f03359bddfe478` and copy its published baseline CSV tables verbatim for the nine models listed in Section 6.1; config checksum verification for XES3G5M is reported there. Concept-prerequisite hyperedges in this paper are formed by enumerating bounded-length paths on the same audited  $E_{\text{pre}}^f$  (Section 4.1).

We extend this notation with a *hyperedge*  $h = (\kappa, \{(\tau_i, e_i)\}_{i=1}^k, f)$ : a kind label  $\kappa \in \{\text{concept-prerequisite, session, discussion-thread, teacher-intervention}\}$ , a set of  $k$  typed member entities  $(\tau_i, e_i)$  with entity type  $\tau_i \in \{\text{student, exercise, concept, teacher, hin}\}$  and a fold identifier  $f$ . A heterogeneous hypergraph  $\mathcal{H}_f = (\mathcal{V}_f, \mathcal{E}_f^{\text{hyper}})$  for fold  $f$  is a set of such hyperedges over the typed node set  $\mathcal{V}_f$ . **Scope of this paper:** only  $\kappa = \text{concept-prerequisite}$  hyperedges (members are concept nodes along audited  $E_{\text{pre}}$  paths) enter Tier 1 training and the quantitative tables below; the other three kinds are future-proof interfaces plus Tier 2 session write-backs at pilot scale (Section 4.1).

### 3.2. Design goal

For each fold  $f$ , the goal is a hypergraph  $\mathcal{H}_f$  constructed using only  $\mathcal{F}_f^{\text{train}}$  evidence, audited for the leakage classes in Table 1 (Section 4.2), that a downstream Tier 1 encoder consumes to predict  $y$  for held-out interactions.

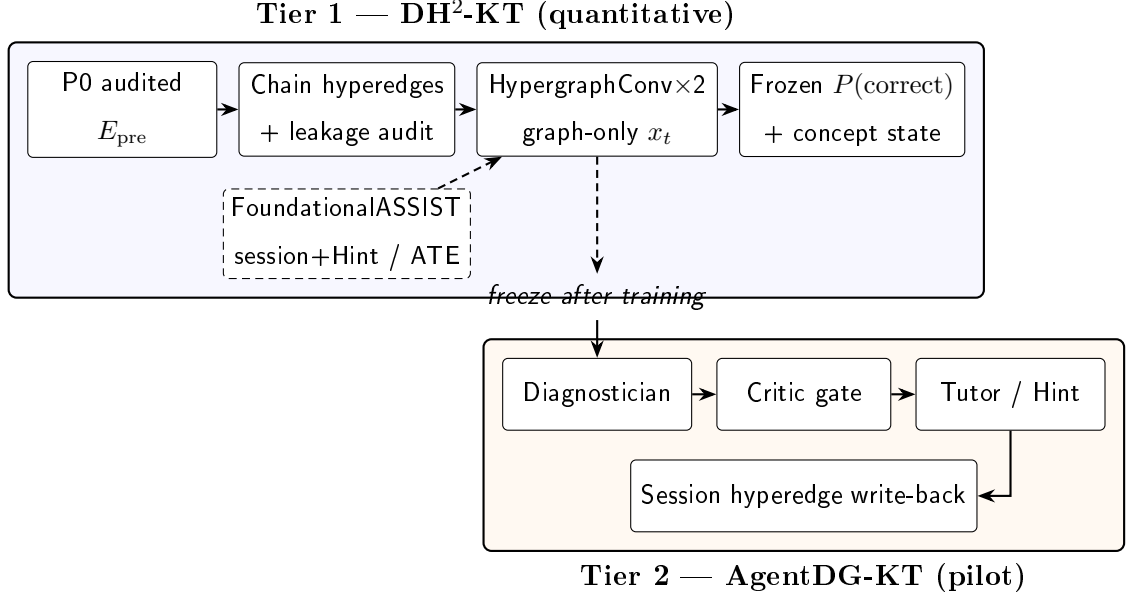


Figure 1: Two-tier DH<sup>2</sup>A-KT architecture. Tier 1 consumes P0’s leakage-audited prerequisite edges as chain hyperedges (solid path; quantitative core on XES3G5M) and optionally FoundationalASSIST session+Hint hyperedges with a limited observational ATE (dashed). After freezing, Tier 2 agents explain and gate those outputs: Diagnostician → Critic → Tutor, with session hyperedge write-back at pilot scale.

## 4. Tier 1: DH<sup>2</sup>-KT

### 4.1. Heterogeneous hypergraph construction

Concept-prerequisite hyperedges are built by enumerating simple directed paths of bounded length on [8]’s audited, acyclic  $E_{\text{pre}}^f$  (Algorithm 1 in code: `build_concept_prerequisite_hyperedges`).

- 1: **Input:** audited, acyclic prerequisite edges  $E_{\text{pre}}^f$  from [8]’s Algorithm 2; chain length bounds  $[\ell_{\min}, \ell_{\max}]$
- 2: **Output:** concept-prerequisite hyperedge set  $\mathcal{E}_f^{\text{cprereq}}$
- 3: Build directed graph  $G$  from  $E_{\text{pre}}^f$

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4: for each root  $r$  with in-degree 0 in  $G$  do
5:   for each simple path  $p$  from  $r$  with  $\ell_{\min} \leq |p| \leq \ell_{\max}$  do
6:     emit hyperedge with members  $\{(\text{concept}, c) : c \in p\}$ 
7:   end for
8: end for

```

Discussion-thread and teacher-intervention hyperedges remain *data-blocked*: no synchronized Forum or Teacher-action logs ship with the public KT corpora we use. Session hyperedges (Student-Exercise-Concept-Hint) are constructed on FoundationalASSIST [9], which provides `hint_count` synchronized to each interaction (Section 7.5). On the three P0 core benchmarks (XES3G5M, ASSISTments 2012, Junyi), session construction stays blocked because those logs lack Hint/Video columns; ES-KT-24 [20] remains an optional Video-bearing source for future work. Tier 1’s primary predictive core on XES3G5M therefore still uses concept-prerequisite hyperedges; FoundationalASSIST supplies the heterogeneous session+Hint evidence and a secondary predictive fold.

#### 4.2. Hyperedge-level leakage audit

[8] audits five leakage classes at the level of pairwise edge provenance (structural, edge, target, temporal, cold-start neighbourhood) using four scalar diagnostics:  $\text{ECR}^{\text{flag}}$ ,  $\text{ECR}^{\text{overlap}}$ ,  $|\rho|$ , and TBMR. These diagnostics are reused unmodified (Section 6.1) on the pairwise projection of a hyperedge set. Table 1 adds three classes that a pairwise audit cannot express because they require a *group* of heterogeneous members, not a single edge.

Measured values on XES3G5M fold 0 appear in Table 2.

#### 4.3. Encoding and hypergraph attention

DH<sup>2</sup>-KT applies two **HypergraphConv** layers per hyperedge kind on concept embeddings, mean-pools across kinds, and masks concepts not participating in any hyperedge. Sequence encoding uses a *graph-only* interaction term  $x_t = \text{proj}(c_t) + \text{proj}(r_t)$  with a learnable graph gain on  $c_t$ , deliberately omitting direct exercise embeddings so predictions cannot bypass the hypergraph (Section 6.3). A dual-gated update combines  $x_t$  with the previous hidden state; a linear head outputs next-step logits. Training adds a graph-sensitivity auxiliary loss encouraging concept states to depend on hypergraph structure.

#### 4.4. Causal-effect layer

We estimate the average treatment effect (ATE) of a binary intervention (e.g., hint given) on next-step correctness via inverse-propensity weighting, following the identification assumption that intervention assignment is unconfounded given observed features – a strong assumption on observational KT data that we report as a limitation rather than a proven property (Section 8.4). Given confounders  $X$ , treatment  $T \in \{0, 1\}$ , and outcome  $Y$ , the propensity score  $\hat{e}(X) = P(T = 1 \mid X)$  is estimated by logistic regression, and

$$\widehat{\text{ATE}} = \frac{\sum_i T_i Y_i / \hat{e}(X_i)}{\sum_i T_i / \hat{e}(X_i)} - \frac{\sum_i (1 - T_i) Y_i / (1 - \hat{e}(X_i))}{\sum_i (1 - T_i) / (1 - \hat{e}(X_i))}. \quad (1)$$

No ATE numbers are reported on the primary XES3G5M benchmark (no hint logs). On FoundationalASSIST we estimate an observational IPW ATE of `hint_usedt` on `correctt+1` with pre- $t$  summaries as confounders (Section 7.5); same-step `discrete_score` is *not* used as the outcome because it is mechanically tied to hint requests in that corpus.

## 5. Tier 2: AgentDG-KT pilot

Tier 2 is deliberately scoped as a pilot, not a benchmark at Tier 1’s quantitative rigor (docs/idea-D-plan.md, following the feasibility analysis in the accompanying project discussion: a full multi-agent LLM re-inference pipeline over millions of interactions is not tractable on a single consumer GPU within a realistic project timeline). Three agents operate on Tier 1’s *frozen* output rather than re-inferring knowledge state:

- **Diagnostician Agent** produces a natural-language rationale for a given Tier-1 prediction, explicitly instructed not to override the given probability.
- **Tutor/Hint Agent** proposes the next hint/exercise and writes a new session hyperedge back into the graph (the agentic closed loop).
- **Critic Agent** is a *mandatory* quality gate that flags any Diagnostician output contradicting or ignoring the Tier-1 number it was meant to explain, before anything is treated as ground truth in a case study.

We ran a 500-sample pilot on XES3G5M fold 0 (seed 42) with Tier 1 frozen (checkpoint from graph-only v2 training) and Qwen2.5-7B (Ollama 0.32.6). The Diagnostician is instructed to end each rationale with a structured **GroundedKCs**: [...] trailer used for ID-anchored KC-Jaccard (Section 7.4); an Independent Generation ablation omits frozen  $P(\text{correct})$  from that prompt while the Critic still receives the true Tier 1 score. Table 3 reports Critic flag rates; v1’s 12.4% flags were predominantly probability *format* mismatches (e.g., “69.3%” vs.  $P=0.693$ ), resolved in v2 by `critic_calibration.py`

and an updated Critic prompt. The Tutor agent wrote 500 session hyperedges back to the graph, closing the agentic loop at pilot scale.

## 6. Experimental setup

### 6.1. Datasets

Following [8]’s benchmark-role assignment, XES3G5M [21] (18,066 learners, 7,652 items, 865 KCs, 6,413,353 interactions) is the *primary* predictive benchmark: low consecutive-KC-repeat rate, non-saturated headline AUC, and the clearest graph-backbone separation among the corpora [8] studied. All DH<sup>2</sup>-KT training and the manipulation check reported below use XES3G5M. ASSISTments 2012 is listed as secondary in the P0 protocol (single-skill Q-matrix); we do *not* reuse P0 ASSISTments 2012 GKT AUC as a comparison column. Instead we report a secondary fold on the newer FoundationalASSIST corpus [9] (Section 7.5), which is *not* interchangeable with classic ASSISTments 2012. Junyi Academy supplies the expert prerequisite DAG for *structural* ground-truth overlap only (Section 7.3); full DH<sup>2</sup>-KT training on Junyi is not run.

### 6.2. Baselines reused from P0

The protocol summary in Section 3.1 underpins reuse. We reuse [8]’s published baseline results (`baselines/p0_reused/baseline_results.csv`, 69 rows, and `baseline_fold_results.csv`, 205 per-fold rows; both copied verbatim from [8]’s `results/tables/`, pinned to commit 5e3d6a01c32a649f1655ac8712f03359bddfe vendored 2026-08-02) for BKT, DKT, AKT, simpleKT, GKT, GIKT, SKT, DyGKT, and DGEKT under train-only, leakage-controlled graph construction, rather than retraining these nine models on a single RTX 3090. Reuse



is valid only if our preprocessing exactly matches [8]’s (same hashed `kc_id` space, same Q-matrix). `scripts/01_verify_p0_preprocessing_match.py` checks that each DH<sup>2</sup>A-KT dataset config points at an unmodified, check-summed copy of [8]’s own config and that the vendored commit is recorded; for XES3G5M this currently confirms the config file matches (sha256 4f9f35d9...d980b2) against the vendored commit above. Processed parquet for XES3G5M, Junyi, and ASSISTments 2012 is present under `external/p0_leakage_audit/data/processed/`. Reviewers may additionally request the companion PDF of [8] as supplementary material while that manuscript is under review.

### 6.3. Manipulation check

Following [8]’s Section 4.7 reasoning, we do not interpret any ablation result as evidence of hypergraph benefit unless DH<sup>2</sup>-KT first passes a manipulation check: AUC must move meaningfully under near-total ( $p = 0.90$ ) destruction of the constructed hyperedge set. A backbone whose AUC does not move under total destruction is *graph-inert*, and its ablation numbers carry no interpretable meaning about graph benefit. On XES3G5M fold 0, DH<sup>2</sup>-KT v2 passes this gate ( $\Delta\text{AUC} = 0.038$ ; Table 5); an earlier graph-inert architecture (pairwise edges + exercise embedding path) failed with  $\Delta\text{AUC} \approx 0.0003$ , motivating the graph-only redesign.

## 7. Results

### 7.1. Predictive performance (RQ1)

Table 4 reports test AUC on XES3G5M under a training budget matched to [8]’s GKT (batch 4, 10 epochs). DH<sup>2</sup>-KT v2 mean AUC is 0.7516 versus

reused GKT 0.8336 and simpleKT 0.8747. We do not claim SOTA predictive performance; mean $\pm$ SD over three folds is  $0.7516\pm0.0015$  vs. GKT  $0.8336\pm0.0010$  (Table 4). The graph-only path is designed so that *our* hypergraph check is interpretable; GKT separately also passes P0’s destruction check (Section 7.2).

### 7.2. Manipulation check (RQ2)

Table 5 and Fig. 2 summarize graph-destruction evidence at the shared P0 anchor ( $p=0.9$  node-drop). All three folds pass:  $\Delta\text{AUC}=0.038/0.043/0.047$  (folds 0/1/2) when frozen weights are re-evaluated on a destroyed hypergraph, confirming hypergraph use under our protocol (an earlier pairwise+exercise-embedding backbone failed this gate,  $\Delta\text{AUC}\approx 0.0003$ ). Training restores the best epoch by train loss; fold 1 previously collapsed on the final epoch and is reported after that safeguard. Reused P0 DDR-downstream results show that **GKT also passes** the same operator/strength on XES3G5M, with a *larger* mean drop ( $0.0877\pm0.0009$ ,  $n=9$  fold-seed runs) after retraining on the destroyed pairwise graph. Protocol differences matter (DH<sup>2</sup>-KT freezes weights; P0 GKT retrains), but either way graph-reliance is *not* unique to DH<sup>2</sup>-KT. simpleKT has no concept-graph adjacency, so  $\Delta\text{AUC}$  under graph destruction is undefined (N/A). Fig. 2 draws an *illustrative* horizontal line at  $\Delta\text{AUC}=0.003$ : roughly an order of magnitude above the observed graph-inert noise floor ( $\approx 3\times 10^{-4}$ ), not a formal hypothesis test from [8]. We treat a backbone as graph-reliant when its destruction  $\Delta\text{AUC}$  clearly exceeds that floor (here 0.038–0.047 across DH<sup>2</sup>-KT folds;  $\approx 0.088$  for GKT).

Table 6 pairs the Tier 1 predictive knobs with Tier 2 faithfulness knobs. Removing the Critic sets the flag rate to 0 by construction while leaving

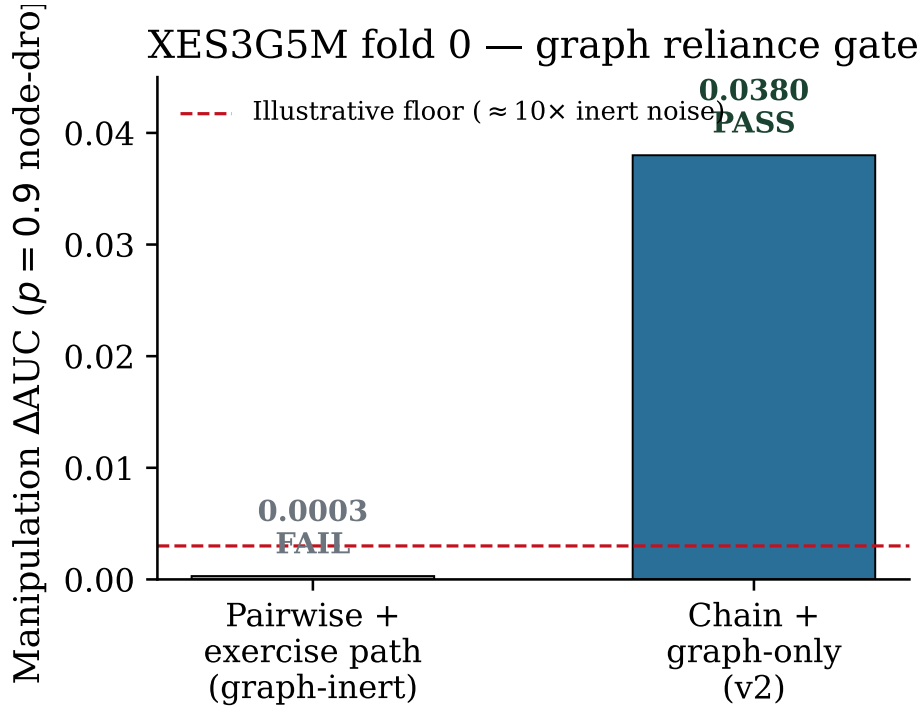


Figure 2: Manipulation-check  $\Delta\text{AUC}$  on XES3G5M fold 0 under  $p=0.9$  node-drop. An early pairwise+exercise-embedding backbone is graph-inert ( $\Delta\text{AUC} \approx 0.0003$ ); the chain+graph-only redesign passes ( $\Delta\text{AUC} = 0.038$ ). The dashed line at 0.003 is an *illustrative* floor ( $\approx 10\times$  the graph-inert noise), not a formal statistical cutoff from [8].

ID-anchored KC-Jaccard unchanged; IG mainly inflates the Critic flag rate ( $0.026 \rightarrow 0.768$ ) with only a small JC shift. On FoundationalASSIST, adding session+Hint hyperedges yields a small AUC gain ( $0.720 \rightarrow 0.722$ ). Dropping  $L_{\text{aux}}$  (`graph_sensitivity_weight` = 0) on fold 0 still yields clean AUC 0.754 and  $\Delta\text{AUC} = 0.039$  (pass), so the graph-only encoder—not the auxiliary sensitivity loss alone—is sufficient for the destruction gate on this split.

### 7.3. Ground-truth structural overlap (Junyi)

Table 7 compares hyperedge-chain overlap with P0’s pairwise  $E_{\text{pre}}$  against the Junyi expert DAG at  $K=|\text{expert}|$ . Edge F1 is 0.165 (hyperedge chain) vs. 0.183 ( $E_{\text{pre}}$ ) on fold 0; folds 1–2 are analogous. The chain representation slightly *underlaps* pairwise  $E_{\text{pre}}$ : collapsing directed paths into multi-node hyperedges, then projecting back to pairs for overlap scoring, can dilute expert-edge recall relative to scoring  $E_{\text{pre}}$  directly. We treat this as structural validation of construction fidelity, *not* as evidence that chain hyperedges improve predictive KT on Junyi — full DH<sup>2</sup>-KT training on Junyi (25M interactions) was not run.

### 7.4. Tier 2 faithfulness pilot

See Table 3 and Section 5. The Tutor wrote 500 session hyperedges at pilot scale. Following [16]’s use of a corpus-wide KC-overlap faithfulness metric and an independent-generation negative control, we additionally compute *ID-anchored KC-Jaccard* over Diagnostician outputs: the model must emit a structured **GroundedKCs**: [...] trailer; support is the target concept, KCs in the recent history window, and their 1-hop neighbors on audited train-only  $E_{\text{pre}}$ . The IG arm uses the same history but omits frozen  $P(\text{correct})$  from the Diagnostician prompt while the Critic still sees the true Tier 1 score (Table 8). On Qwen2.5-7B (Ollama,  $n=500$ ), grounded explanations have flag rate 0.026 and mean JC 0.152; IG raises the Critic flag rate to 0.768 while mean JC stays near 0.142 ( $\Delta\text{JC}\approx +0.010$ ), so the gate mainly catches probability-ungrounded rationales rather than rewriting KC overlap. StubLLM harness rows ( $n=20$ ) validate the pipeline (flag 0.0 vs. 1.0 at matched JC). From the calibrated Ollama v2 run we drew a fixed 40-sample

pack (seed 42) for two independent raters (A1, B1) on Likert faithfulness and usefulness (1–5). Pooled means are  $4.01 \pm 1.07$  (faithfulness) and  $3.53 \pm 0.71$  (usefulness); pairwise Spearman  $\rho = 0.71 / 0.58$  and within- $\pm 1$  rates of 92.5% / 97.5% (Table 9). A third rater was optional and not required for this pilot report.

#### 7.5. Secondary corpus: FoundationalASSIST (*session+Hint*)

FoundationalASSIST [9] provides  $\approx 1.74\text{M}$  skill-linked interactions with synchronized `hint_count` (hint-use rate 5.2%). Using the same learner-temporal protocol as P0 (0.7/0.1/0.2, fold 0), we infer train-only  $E_{\text{pre}}$  (280 edges after cycle pruning), build 5,840 concept-prerequisite chain hyperedges, and construct 308,670 session hyperedges on train users (11.6% with a Hint member;  $\text{ECR}^{\text{flag}}=0$ , group-membership leak=0). DH<sup>2</sup>-KT fold 0 AUC is 0.720 with concept-prerequisite hyperedges alone ( $n=298,500$  predictions). Adding up to 20k multi-concept session hyperedges (Hint-preferring selection; concept co-occurrence projection) raises AUC to 0.722 ( $\Delta+0.002$ ) — a small positive shift under the same local budget, not a matched GKT comparison. Using the causal layer on train transitions, the IPW ATE of hint use on *next-step* correctness is  $-0.207$  (naive mean difference  $-0.255$ ; trimmed IPW  $-0.211$ ; bootstrap 95% CI  $[-0.230, -0.187]$  on 50k-row re-samples;  $n_{\text{treated}}=36,151$ ). Propensity scores hit the clip floor ( $[0.05, 0.219]$ ), indicating weak overlap: struggling learners disproportionately request hints, so the negative association is *not* interpretable as a proven causal penalty of giving hints. Table 10 summarizes.

## 8. Discussion

### 8.1. Threats to validity

Our XES3G5M comparison uses [8]’s reused baselines under matched batch/epoch budget (Table 4); preprocessing configs are pinned to P0’s vendored commit. DH<sup>2</sup>-KT mean AUC is  $0.7516 \pm 0.0015$  vs. GKT  $0.8336 \pm 0.0010$  (paired  $\Delta = -0.082 \pm 0.003$  over three folds). Both DH<sup>2</sup>-KT and GKT pass strong graph-destruction checks; the AUC gap therefore cannot be read as “DH<sup>2</sup>-KT uniquely buys graph-reliance.” Section 8.2 repositions value around audit, heterogeneous/causal interfaces, and Tier 2 rather than exclusive graph sensitivity. Primary claims remain scoped to concept-prerequisite hyperedges on XES3G5M; FoundationalASSIST is secondary (Section 7.5). Passing destruction confirms graph use for a backbone, not that a specific hyperedge kind drives predictive gain.

### 8.2. What justifies DH<sup>2</sup>-KT if graph-reliance is not unique?

Because GKT also passes destruction while beating DH<sup>2</sup>-KT on clean AUC (Table 5), the remaining case for DH<sup>2</sup>-KT is what reused GKT does *not* provide: (i) hyperedge-level leakage diagnostics beyond pairwise  $E_{\text{pre}}$  (Table 1); (ii) a shared schema for heterogeneous members (session/Hint/teacher/forum/video), exercised for session+Hint on FoundationalASSIST while discussion-thread and teacher types stay data-blocked; (iii) an inverse-propensity ATE layer for pedagogical interventions (Section 4.4; limited observational hint ATE on FoundationalASSIST); and (iv) a Critic-gated explanation loop over frozen  $P(\text{correct})$  (Section 5). Protocol note: DH<sup>2</sup>-KT  $\Delta\text{AUC}$  is frozen-weight re-evaluation; P0 GKT  $\Delta\text{AUC}$  retrains on the destroyed pairwise

graph — complementary, not apple-to-apple. **Prefer high-AUC baselines** when ranking alone is the product goal; **prefer DH<sup>2</sup>-KT** when stakeholders need higher-order auditability, an extensible heterogeneous/causal interface, and/or Critic-gated explanations, accepting the  $\approx 8$  point AUC gap versus GKT ( $-0.082 \pm 0.003$ ; Table 4) as the cost of our graph-only path — not a claim that GKT is graph-inert.

### 8.3. Tier 2 faithfulness

Tier 2’s 12.4% v1 flag rate was largely a Critic prompt artifact (% vs. decimal); calibration reduced it to 0% on 500 samples without changing Tier 1 numbers. Dual-rater scores (Table 9) place faithfulness near “aligned” ( $\approx 4/5$ ) and usefulness in the “somewhat–mostly useful” band ( $\approx 3.5/5$ ), with high within- $\pm 1$  agreement; usefulness Spearman is moderate, reflecting more subjective teacher-facing judgments.

### 8.4. Limitations

- Discussion-thread and teacher-intervention hyperedges remain data-blocked; session+Hint hyperedges are demonstrated on FoundationalASSIST (concept projection;  $+0.002$  AUC vs. concept-only) but are not the primary XES3G5M training graph.
- The causal-effect layer’s identification assumption (no unmeasured confounding given observed features) is stated, not proven; the FoundationalASSIST hint ATE shows weak propensity overlap and should not be read as a recommendation against hints.

- The hyperedge leakage audit’s pairwise-projection approximation (Section 4.2) weights all projected pairs equally; a hyperedge-native statistic is future work.
- Tier 2 pilot: one fold, one local 7B LLM; Critic sensitivity to response format required calibration; human ratings use  $n=40$  and two raters (no third rater; not a classroom deployment study).
- Reproduction environment: Windows workstation, single RTX 3090; 42/42 unit tests pass in the released codebase.

## 9. Conclusion and future work

We presented DH<sup>2</sup>A-KT, a two-tier framework combining leakage-controlled hypergraph knowledge tracing (DH<sup>2</sup>-KT) with a frozen-core LLM agent pilot. On XES3G5M, graph-only DH<sup>2</sup>-KT and reused GKT both pass strong destruction checks; DH<sup>2</sup>-KT remains below GKT AUC ( $0.752\pm0.002$  vs.  $0.834\pm0.001$ ) under a graph-only interaction path. Quantitative claims are scoped to concept-prerequisite chain hyperedges. On Junyi, those hyperedges show structural overlap with the expert DAG (edge-F1  $\approx 0.17$ ) without full-model training. Tier 2 demonstrates a Critic-gated loop at 500-sample scale with 0% flag rate after calibration and dual-rater Likert means of  $\approx 4.0$  /  $3.5$  for faithfulness / usefulness. On FoundationalASSIST we add session+Hint hyperedges, a secondary fold-0 AUC, and a limited observational hint ATE with propensity diagnostics. Future work includes richer confounders / sensitivity analysis for ATE, a third rater or larger human sample, ES-KT-24 Video wiring, and hyperedge-native leakage statistics.



## **CRedit authorship contribution statement**

**Tuan Dao Minh:** Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Visualization. **Khanh-Trinh Nguyen:** Validation, Data curation, Writing – review & editing. **Duong Nguyen Tien:** Validation, Data curation, Writing – review & editing. **Quoc Khanh Ngo:** Investigation, Resources, Writing – review & editing. **Van-Hau Nguyen:** Validation, Resources. **Le Hoang Son:** Conceptualization, Supervision, Project administration, Funding acquisition, Writing – review & editing.

## **Declaration of competing interest**

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## **Data availability**

Code for DH<sup>2</sup>A-KT will be made available on request. P0’s audited prerequisite edges and reused baseline results are vendored under [8]’s own release terms (pinned commit, Section 6.2); FoundationalASSIST [9] must be obtained directly from Hugging Face under its Responsible Use Guidelines and cannot be redistributed by the authors.

## **Acknowledgment**

All Tier 1 experiments in this paper are designed to run on a single consumer-grade NVIDIA RTX 3090 (24GB), matching the compute budget

already validated by [8] on the same primary benchmarks. FoundationalAS-SIST [9] was accessed under the Hugging Face dataset’s Responsible Use Guidelines for this study.

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Table 1: Hyperedge-level leakage taxonomy (extends [8] Table 1)

| Class               | Source                                                                                                           | Diagnostic                          |
|---------------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| Group-membership    | A hyperedge with $\geq 1$ member drawn from a held-out interaction while other members are train-only            | compute_group_membership_leakage    |
| Cross-modal         | A Forum/Video embedding pooled over content timestamps after the prediction cutoff                               | compute_cross_modal_leakage         |
| Intervention-timing | A Teacher/Hint hyperedge whose content was effectively chosen using knowledge of the student's later correctness | compute_intervention_timing_leakage |

Table 2: Concept-prerequisite hyperedge audit on XES3G5M fold 0. Pairwise TBMR/ $|\rho|$  are skipped when the pairwise projection exceeds the implementation cap of 100,000 pairs (11,568,053 projected pairs in this fold; see audit notes).

| Metric                     | Value   |
|----------------------------|---------|
| Hyperedges                 | 446,305 |
| ECR <sup>flag</sup>        | 0.0     |
| Group-membership leak rate | 0.0     |

Table 3: Tier 2 agent faithfulness pilot (XES3G5M fold 0,  $n=500$ , seed 42, Qwen2.5-7B via Ollama). Flag rate = fraction of Diagnostician outputs flagged by the mandatory Critic. v2 applies probability-format calibration (`critic_calibration.py`).

| Run           | Backend    | Flag rate            | Mean $P(\text{correct})$ |
|---------------|------------|----------------------|--------------------------|
| Stub (sanity) | StubLLM    | 0.000 (0/500)        | 0.742                    |
| Ollama v1     | qwen2.5:7b | 0.124 (62/500)       | 0.746                    |
| Ollama v2     | qwen2.5:7b | <b>0.000</b> (0/500) | 0.753                    |



Table 4: Test AUC on XES3G5M under matched training budget (batch 4, 10 epochs, max seq. len. 200). P0 fold AUCs from `baselines/p0_reused` (train-only graphs); DH<sup>2</sup>-KT v2 uses chain concept-prerequisite hyperedges and graph-only interaction. Mean $\pm$ SD over three learner folds.

| Fold                             | DH <sup>2</sup> -KT v2              | GKT (P0)                            | simpleKT (P0)                       |
|----------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
| 0                                | 0.7501                              | 0.8346                              | 0.8744                              |
| 1                                | 0.7518                              | 0.8338                              | 0.8747                              |
| 2                                | 0.7531                              | 0.8326                              | 0.8750                              |
| <b>Mean<math>\pm</math>SD</b>    | <b>0.7516<math>\pm</math>0.0015</b> | <b>0.8336<math>\pm</math>0.0010</b> | <b>0.8747<math>\pm</math>0.0003</b> |
| $\Delta$ vs. GKT (mean $\pm$ SD) |                                     | $-0.0820\pm0.0025$                  |                                     |

Table 5: Manipulation check at node-drop  $p=0.9$  on XES3G5M. DH<sup>2</sup>-KT: train on clean hyperedges, re-evaluate frozen weights on destroyed hyperedges (folds 0–2). Training restores the best epoch by train loss (fold 1 previously collapsed on the final epoch). GKT: P0 DDR-downstream mean  $\Delta$ AUC over  $n=9$  fold–seed runs ([8]; retrain on destroyed pairwise graph). simpleKT has no concept-graph input, so graph destruction is undefined.

| Model                           | AUC clean | $\Delta$ AUC ( $p=0.9$ ) | DDR   | Pass? |
|---------------------------------|-----------|--------------------------|-------|-------|
| DH <sup>2</sup> -KT v2 (fold 0) | 0.753     | 0.0380                   | 0.990 | Yes   |
| DH <sup>2</sup> -KT v2 (fold 1) | 0.749     | 0.0432                   | 0.990 | Yes   |
| DH <sup>2</sup> -KT v2 (fold 2) | 0.753     | 0.0469                   | 0.990 | Yes   |
| GKT (P0, mean $\pm$ SD)         | —         | 0.0877 $\pm$ 0.0009      | 0.990 | Yes   |
| simpleKT (P0)                   | —         | N/A                      | —     | N/A   |

Table 6: Paired ablation of DH<sup>2</sup>A-KT components. Tier 1 metrics are on XES3G5M unless noted; Tier 2 faithfulness uses the fold 0 frozen checkpoint and Qwen2.5-7B (Ollama,  $n=500$ ) unless noted. “w/o Critic” disables the gate on the same grounded Diagnostician outputs (flag rate 0 by construction; JC unchanged). Independent generation (IG) strips frozen  $P(\text{correct})$  from the Diagnostician. Session hyperedges are FoundationalASSIST-only. “w/o  $L_{\text{aux}}$ ” sets `graph_sensitivity_weight` = 0 (fold 0 retrain).

| Setting                       | AUC               | Manip. $\Delta$ AUC | Flag  | Mean JC |
|-------------------------------|-------------------|---------------------|-------|---------|
| Full (grounded + Critic)      | 0.752 $\pm$ 0.002 | 0.038–0.047         | 0.026 | 0.152   |
| IG (w/o $P$ in Diag.)         | —                 | —                   | 0.768 | 0.142   |
| w/o Critic                    | —                 | —                   | 0.000 | 0.152   |
| FA: concept-only              | 0.720             | —                   | —     | —       |
| FA: + session HE              | 0.722             | —                   | —     | —       |
| w/o $L_{\text{aux}}$ (fold 0) | 0.754             | 0.039               | —     | —       |

Table 7: Ground-truth cross-validation on Junyi Academy (fold 0, top- $K=|\text{expert}|=1131$ ). Hyperedge chain slightly underlaps pairwise  $E_{\text{pre}}$  but remains the same order of magnitude.

| Representation      | Edge F1 | Edge prec. | Edge rec. |
|---------------------|---------|------------|-----------|
| P0 $E_{\text{pre}}$ | 0.183   | 0.183      | 0.183     |
| Hyperedge chain     | 0.165   | 0.165      | 0.165     |

Table 8: Tier 2 explanation faithfulness: grounded Diagnostician vs. independent generation (IG). ID-anchored KC-Jaccard (JC) measures overlap between **GroundedKCs** trailer IDs and the support set  $\{\text{target}\} \cup \text{history} \cup \text{1-hop } E_{\text{pre}}$ . IG omits frozen  $P(\text{correct})$  from the Diagnostician prompt; the Critic still sees the true Tier-1 score.

| Arm      | Backend             | $n$ | Flag rate | Mean JC |
|----------|---------------------|-----|-----------|---------|
| Grounded | StubLLM (harness)   | 20  | 0.000     | 0.191   |
| IG       | StubLLM (harness)   | 20  | 1.000     | 0.191   |
| Grounded | qwen2.5:7b (Ollama) | 500 | 0.026     | 0.152   |
| IG       | qwen2.5:7b (Ollama) | 500 | 0.768     | 0.142   |

Table 9: Human rating of Tier 2 outputs (XES3G5M fold 0 Ollama v2 pilot subsample,  $n=40$  explanations, 2 independent raters, Likert 1–5). Agreement is pairwise Spearman  $\rho$  and fraction of scores within  $\pm 1$ .

| Scale        | Mean $\pm$ SD   | Agreement (A1 vs. B1)                |
|--------------|-----------------|--------------------------------------|
| Faithfulness | $4.01 \pm 1.07$ | $\rho=0.71$ ; within $\pm 1$ : 92.5% |
| Usefulness   | $3.53 \pm 0.71$ | $\rho=0.58$ ; within $\pm 1$ : 97.5% |

Table 10: FoundationalASSIST secondary corpus (fold 0). Session hyperedges use a 30-minute gap; Hint members appear when `hint_count` > 0. Predictive AUC is observational (no P0-reused GKT/simpleKT column). Hint ATE is IPW of `hint_usedt` on `correctt+1` under pre- $t$  confounders; propensity mass hits the clip floor — treat as limited observational evidence.

| Quantity                                                  | Value                      |
|-----------------------------------------------------------|----------------------------|
| Interactions (canonical)                                  | 1,740,987                  |
| Learners / items / KCs                                    | 5,000 / 3,359 / 209        |
| Hint-use rate                                             | 0.052                      |
| Train-only $E_{\text{pre}}$ edges (after cycle prune)     | 280                        |
| Concept-prerequisite chain hyperedges                     | 5,840                      |
| Session hyperedges (train users)                          | 308,670                    |
| with $\geq 1$ Hint member                                 | 35,752 (11.6%)             |
| Session hyperedge $\text{ECR}^{\text{flag}}$ / group leak | 0 / 0                      |
| DH <sup>2</sup> -KT fold 0 AUC (concept-prereq)           | 0.720                      |
| + session HE (20k, Hint-prefer)                           | 0.722 ( $\Delta + 0.002$ ) |
| $n$ next-step predictions                                 | 298,500                    |
| Hint ATE (IPW) / naive mean diff                          | -0.207 / -0.255            |
| trimmed IPW (propensity 10–90%)                           | -0.211                     |
| bootstrap 95% CI (subsample)                              | [-0.230, -0.187]           |
| $n$ treated / control (train transitions)                 | 36,151 / 660,321           |
| propensity [min, max]                                     | [0.05, 0.219]              |